STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 5: Work, Energy and Power

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. When is work zero with displacement? [1]
2. Kinetic energy? [1]
3. SI unit of power? [1]
4. Work-energy theorem? [1]
5. Spring energy? [1]
6. What is conserved in an isolated elastic collision? [1]
7. Near-earth gravitational potential energy? [1]
8. Work by centripetal force in uniform circular motion? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. 10 N over 4 m along force: work? [2]
10. 3 kg at 4 m/s: K? [2]
11. 100 W for 20 s: energy? [2]
12. 200 N perpendicular to 5 m displacement: work? [2]
13. k=200 N/m, spring stretched .1 m: energy? [2]
14. 1 kg cart at 5 m/s sticks to stationary 4 kg: speed? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. 2 kg speeds from 2 to 6 m/s: net work? [3]
16. 5 kg falls 3 m, g=10: final speed? [3]
17. F=4x N from x=0 to 3 m: work? [3]
18. 1000 kg car at 20 m/s stops in 50 m: W and force magnitude? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. 2 kg drops 5 m frictionlessly, g=10: speed; with 20 J loss? [4]
20. 1 kg cart at 8 m/s sticks to 3 kg stationary: speed and energy loss? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations worth knowing

- Work: $W=Fs\cos(\theta)$; if force is perpendicular, $W=0$.
- Kinetic energy $K=(1/2)mv^2$; gravitational potential near Earth $U=mgh$; spring energy $U_s=(1/2)kx^2$.
- Work-energy theorem: $W_{net}=\Delta K=(1/2)m(v^2-u^2)$.
- Power $P=W/t$; energy transferred $E=Pt$.
- Momentum conservation in a perfectly inelastic collision: $m_1u_1+m_2u_2=(m_1+m_2)v$.
- For $F=kx$ from $x=0$ to $x=a$: $W=\int F\,dx=(1/2)ka^2$.
- For a falling body without losses: $mgh=(1/2)mv^2$, so $v=\sqrt{2gh}$.
- Stopping force magnitude for constant force: $|F|=|W|/\text{distance}$.

Use each equation only when its assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Solutions are grouped here after the question paper and formula reference.

Question 1

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: When force is perpendicular to displacement.
Final answer: When force is perpendicular to displacement.

Question 2

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: mv^2.
Final answer: mv^2.

Question 3

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Watt.
Final answer: Watt.

Question 4

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Net work equals change in kinetic energy.
Final answer: Net work equals change in kinetic energy.

Question 5

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: kx^2.
Final answer: kx^2.

Question 6

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Momentum and kinetic energy.
Final answer: Momentum and kinetic energy.

Question 7

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: mgh.
Final answer: mgh.

Question 8

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Zero.
Final answer: Zero.

Question 9

Working: Work W=Fs cos0=10(4)(1)=40 J.
Final answer: 40 J.

Question 10

Working: K=(1/2)mv^2=0.5(3)(4^2)=1.5(16)=24 J.
Final answer: 24 J.

Question 11

Working: Energy transferred E=Pt=100(20)=2000 J.
Final answer: 2000 J.

Question 12

Working: W=Fs cos90 deg=0 J because the force is perpendicular to displacement.
Final answer: 0 J.

Question 13

Working: Spring energy U=(1/2)kx^2=0.5(200)(0.1^2)=100(0.01)=1 J.
Final answer: 1 J.

Question 14

Working: Conserve momentum: (1)(5)+(4)(0)=(1+4)v, so v=5/5=1 m/s.
Final answer: 1 m/s.

Question 15

Working: Net work=change in kinetic energy=0.5m(v^2-u^2)=0.5(2)(36-4)=32 J.

Final answer: 32 J.

Question 16

Working: mgh=(1/2)mv^2; mass cancels, so v=sqrt(2gh)=sqrt(2(10)(3))=sqrt60=7.75 m/s.

Final answer: sqrt60approx 7.75 m/s.

Question 17

Working: F=4x. Work is the area under F-x graph: integral from 0 to 3 of 4x dx=[2x^2]_0^3=18 J.

Final answer: 18 J.

Question 18

Working: Initial K=0.5(1000)(20^2)=200,000 J. Work by braking=-200,000 J=-200 kJ. Average stopping force magnitude=200,000/50=4000 N.

Final answer: -200 kJ; 4000 N.

Question 19

Working: Without loss, mgh=K gives v=sqrt(2gh)=sqrt(100)=10 m/s. If 20 J is lost, initial potential energy=2(10)(5)=100 J and final K=80 J; v=sqrt(2K/m)=sqrt80=8.94 m/s.

Final answer: 10 m/s; sqrt80approx 8.94 m/s.

Question 20

Working: Momentum conservation gives (1)(8)=(1+3)v, so v=2 m/s. Initial K=0.5(1)(64)=32 J; final K=0.5(4)(2^2)=8 J; energy lost=32-8=24 J.

Final answer: 2 m/s; 24 J lost.